Use Case Specification: UC-04

- **ID / Code:** UC-04
- **Use Case Name:** View Invoices & Add Comments
- **Description:** Allows a user to view, download and track the status of invoices belonging to their assigned clients, and to leave contextual comments on a specific invoice, mentioning other internal users (UR-1.2, FR-2.8, FR-6.8).
- **Primary Actor:** Base User (inherited by Administrator and Super-Administrator)
- **Scope:** Subscription Management & Invoicing System
- **Level:** User Goal
- **Stakeholders & Interests:**
  - *Base User:* Wants to check the lifecycle status of invoices and coordinate with colleagues on a specific invoice (UR-1.2, FR-2.8, FR-6.8).
  - *Mentioned user:* Wants to be notified only when their attention is required on a specific invoice.
  - *Company (Product Owner):* Wants collaboration to stay contextual and traceable, avoiding unstructured chat (FR-2.8).
- **Preconditions:** The user is authenticated (UC-01) and has permission to access the invoice's customer (FR-6.8); attempting to access an unauthorized invoice is rejected (FR-6.8, NFR-5.6).
- **Postconditions:** If a comment is added, it is stored linked to the invoice, with author and timestamp, and mentioned users are notified.
- **Success Guarantee:** The user views the invoice with its current status (FR-2.4) and, optionally, downloads its PDF (FR-6.8) or publishes a comment that notifies the mentioned users.
- **Minimal Guarantee:** The invoice data is never modified through this use case; comments that fail to save are not partially stored; no invoice outside the user's authorized scope is ever accessible, even via direct URL (FR-6.8, NFR-5.6).
- **Trigger:** The user selects an invoice from the dashboard or a search result (UC-03).
- **Main Success Scenario:**

1. **User:** Selects an invoice.
2. **System:** Displays the invoice details: number, customer, items, taxes, amounts, due date, current status ("draft", "ready to issue", "issued", "sent", "overdue", "paid", "partially reconciled") (FR-2.4), attachments (FR-2.2.3) and its comment thread.
3. **User:** Writes a comment on the invoice and mentions one or more internal users using "@".
4. **System:** Validates the comment and that the mentioned users exist and are active.
5. **System:** Stores the comment linked to the invoice, with author and timestamp.
6. **System:** Sends a targeted notification to each mentioned user via the notification subsystem (FR-2.8, UC-23).
7. **System:** Displays the updated comment thread.

- **Extensions:**
  - 2a. User downloads the invoice (FR-6.8):
    - 2a1. User requests the PDF.

- 2a2. System generates/serves the PDF preserving special characters correctly (NFR-2.6). The use case ends.

- 3a. User only views the invoice and leaves:
  - 3a1. The use case ends with no changes.
- 4a. Mentioned user does not exist or is inactive:
  - 4a1. System informs the user and highlights the invalid mention.
  - 4a2. User corrects the mention; the flow resumes at step 3.
- 4b. Empty comment:
  - 4b1. System does not allow the comment to be published.
- 6a. Notification cannot be delivered:
  - 6a1. System keeps the comment, queues the notification for retry and shows the mention in the recipient's dashboard (FR-6.2).